ECLIPS Internal Workshop 2

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Preliminary Results

Phase I: Recruiting / Study Participation Decision

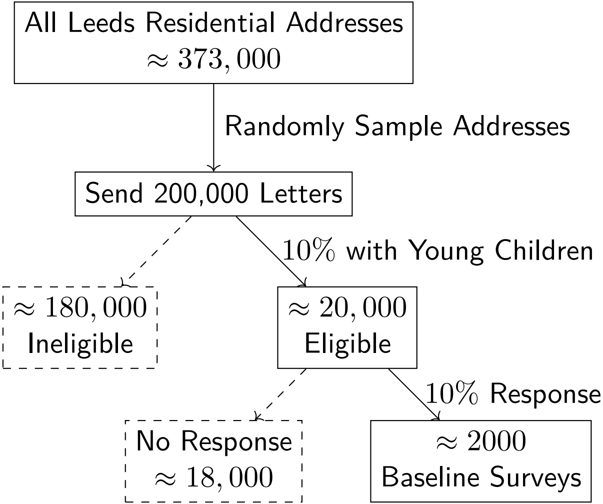
Make a simplified flowchart: Frank/Ludovica birds-eye view of Phase I and Phase II. Then "zoom in" on specific parts and fill in details. Could be more like Angela's flowchart initial boxes. E.g. addresses, letters, randomization, re-balancing, dust kits, feedback

1. Sampling design
2. Recruitment / Intake (don't mention community pathway in this talk!)
3. Kit allocation & Randomization & Exposure Survey

- *These are the things we will analyze today; later we may look at dust / feedback surveys*
- Start with super simple 3 blocks flowchart
- As we progress through the talk give some more detail; possibly show more flowcharts with specifics (or not)

Phase I: Recruiting / Study Participation Decision

Update this by removing ineligible / eligible and putting actual responses



5 GBP gift card for sign-up and baseline survey completion

Research Question: who responds and why?

- Various kinds of responses: surveys, returning kits, experience surveys etc.
- Design of recruiting materials: RCT
- Design of the kits: RCT
- Demographic characteristics (don't observe at the individual level but do observe at LSOA); do they look like the population of Leeds overall?
- Can look at this for each kind of response

Sampling Design: Data Sources

Leeds Council Tax Data:

- Source of residential addresses in Leeds
- 376,205 rows: one per chargeable dwelling
- Columns we use: Address 1-4, Unique Property Reference Number (UPRN)

ONS address data:

Merge Postcode, OA, LSOA and MSOA with Council Tax data (by UPRN)

Energy Performance Certificates (EPC):

Merge Housing age (use latest certificate) with Council Tax data (by UPRN)

Census Data (2021) at LSOA level

Merge LSOA-level demographics with Council Tax data (by LSOA)

Exclusion / De-duplication

See appendix slide for details.

Stratified Random Sampling to Improve Precision

How to allocate a total of N letters across LSOAs / housing age?

Eligibility

- Letters sent to households without a child aged 1-6 are wasted
- Approximate # eligible from 2021 Census: 0-4 & 5-9 bands
- $E_\ell =$ (# Eligible in LSOA ℓ) / (Total # Eligible)
- Send $N \times E_\ell$ letters to LSOA ℓ (Hamilton allocation)

Housing Age

- $O_\ell =$ Share of Old houses in LSOA ℓ
- Suppose N_ℓ letters will go to LSOA ℓ based on eligibility.
- Send $N_\ell \times O_\ell$ of the N_ℓ letters go to Old houses
- The rest go to NonOld houses (Hamilton allocation)

Map of where we sent letters

- Map of where we sent letters for both rounds; possibly next to or alongside a plot of the estimated share eligible across LSOAs.

Randomized Controlled Trial #1: Recruiting Letter

Generalized Risk (G)

The UK Health Security Agency says that lead can be found in things like: Paint on old walls, doors, windowsills, radiators, furniture, toys, and ornaments; Old plumbing, including pipes, tanks, and taps; Some spices brought from other countries; Some traditional medicines and beauty products (like surma or kohl eye makeup); Certain cooking pots (including grey pewter pots) from countries outside the UK and Europe.

Personalized Risk (P)

We see from local records that your home was built before 1967, so it may have old paint or pipes containing lead. Finding out if your child has come into contact with lead could help protect their health.

Control letter contains neither; Personalized Risk letter contains both

Elevated Childhood Lead Interagency Prevalence Study

The Occupier
<<addr1>>
<<addr2>>
<<addr3>>
<<addr4>>
<<POSTCODE>>

<<DATE>>

Dear Resident,

We invite you to **join an important health study** that could help protect children's health. We are ECLIPS (Elevated Childhood Lead Interagency Prevalence Study), a UK team of researchers and practitioners. This study is about lead, a toxic metal that can still be found in many homes and some everyday products. Even small amounts of lead can seriously harm children's health and cause learning difficulties. We picked your address at random with 30,000 others to help us learn how children might come into contact with lead in their homes and neighbourhoods.

The UK Health Security Agency says that lead can be found in things like:

- Paint on old walls, doors, windowsills, radiators, furniture, toys, and ornaments
- Old plumbing, including pipes, tanks, and taps
- Some spices brought from other countries
- Some traditional medicines and beauty products (like surma or kohl eye makeup)
- Certain cooking pots (including grey pewter pots) from countries outside the UK and Europe.

We see from local records that your home was built **before 1967**, so it may have old paint or pipes containing lead. Finding out if your child has come into contact with lead could help protect their health.

How can you help?

If you live with a **child aged 1-6 years old**, you can fill in our 5-minute online survey following this QR code. Upon completion, you will be eligible to receive a £5 gift card to thank you for your help. Your information will be kept private and secure. The QR code can be used only once.

The survey is the first step in our study. Based on your answers, we may invite you to join the next part of our study. If selected, you'll receive a kit in the post to do a simple finger-prick test on your child at home to check for lead. You can receive up to **£45** in total for taking part.

Need help or have questions?

Scan to start the survey:

qr_code

Email: pb_aware@northumbria.ac.uk
Website: leadsafefutures.org
Translations, an audio version, and other accessibility options are available on our website

Thank you for helping us create lead-safe futures for children across the UK.
Kind regards.

[Please turn over for more information about our study →]

ECLIPS_OutreachLetter_T2_v1.0 16.10.25 NU Ethics 10258

Elevated Childhood Lead Interagency Prevalence Study

Why is this important?

Lead can still be found in many places and products around us. It causes learning difficulties, attention and behaviour problems, and long-term health issues. But right now, we don't have good information about how many UK children are harmed by lead each year. Studies estimate that it may be around 1 in every 60 children – enough children to fill Wembley Stadium two times over – but the true number could be even higher. We need to gather better data to learn which children are at risk. Without this information, lead exposure can go unnoticed until it's too late to address the impact.

Who are we?

We are ECLIPS (Elevated Childhood Lead Interagency Prevalence Study), a UK research team of child health experts, environmental scientists, and policy researchers working together to help keep children healthy. We have representatives from Northumbria University, UKHSA, HSE, Leeds Teaching Hospital NHS Trust, University of Oxford, Warwick University, University of Bristol, LEAPP Alliance, Synnovis, and King's College London.

How can you participate in this study?

Does a child aged **1-6 years** live in your home? If yes, you are eligible to participate. After you complete the survey, if we pick you to take part, you may:

1. **Collect a blood spot sample**
   - You will receive a sampling kit to collect a few drops of your child's blood. It includes instructions, a lancet, wipes, plaster, collection card, and prepaid return envelope.
2. **Complete two online surveys**
   - A 20–30-minute questionnaire about your child, home, and lifestyle, available with the blood sampling kit.
   - A 5-minute feedback survey, available when receiving results.
3. **Send soil and dust samples**
   - You may receive a sampling kit to collect household dust (from your vacuum) and soil from your garden or backyard to measure lead levels.

You will receive your child's results in a letter from our qualified health team along with a guide on how to reduce lead exposure. If any follow-up is needed, we will guide you through the next steps.

What happens to my information?

Your information will be kept strictly confidential, stored securely, and only accessible to authorised researchers. Data may be used for future research with ethical approval and your consent.

Can I change my mind about taking part in this study?

Taking part in this survey and the full study is voluntary and you can change your mind at any time without giving any reason. If consent is withdrawn, any personal data collected up to that point will be destroyed unless consent is given to keep it.

pb_aware@northumbria.ac.uk | www.leadsafefutures.org

Randomization of Letter Types

Recall: Randomization: "personalized" treatment only for Old homes

- Old stratum: $1/3$ control, $1/3$ generalized, $1/3$ personalized
- NonOld stratum: $1/2$ control, $1/2$ generalized

Kit Allocation

Flag as "disadvantaged" if *any* of the following:

- Low income: $\leq$ £24,999
- Low education: "No formal qualifications" or "Education up to age 16"
- Language: English non-native *and* reading/writing "Quite hard" / "Very hard"

Re-balancing

- Budgeted for 500 kits and associated lab capacity
- Expect fewer responses from some groups, so recruit more participants than kits
- Earmark 50% of kits for "disadvantaged"; choose randomly among them
- Allocate remaining kits randomly among not disadvantaged

Randomize Kit Types

- **I can't remember what's different about the two kits!**
- 50-50 random assignment *stratified* by (Letter treatment) $\times$ (Disadvantaged)

Map of where we sent kits?

- Haven't looked at this yet; might look quite different because of re-balancing
- Also should say *who* got kits. I think we ended up sending everyone who was disadvantaged a kit.
- This relates to the question about how the demographics of our respondents match (or do not) with the population of Leeds

Adaptive Design

Budget for 150k letters and 2k baseline surveys (5 GBP gift card)

Round 1

Send 30k letters; wait 45 days; # of responses R_1 determines letters sent in Round 2

Round 2

- Plug-in response rate $\hat{p} = R_1/30{,}000$; send enough to reach 2,000:

$$\text{send} = \min\Big(120\text{k},\ 10\text{k} \times \big\lceil (2000 - R_1)\,/\,(10{,}000\,\hat{p}) \big\rceil\Big)$$

- Round **up** to next 10k; stop early if $R_1 \geq 2{,}000$ (target met) or $R_1 < 20$ (disaster)

Kit Allocation

- Split 500 kits across rounds $\propto$ expected responses; Round 2 gets remainder
- Same rebalancing rule in each round; selection probabilities are **round-specific** (depends on who responds)

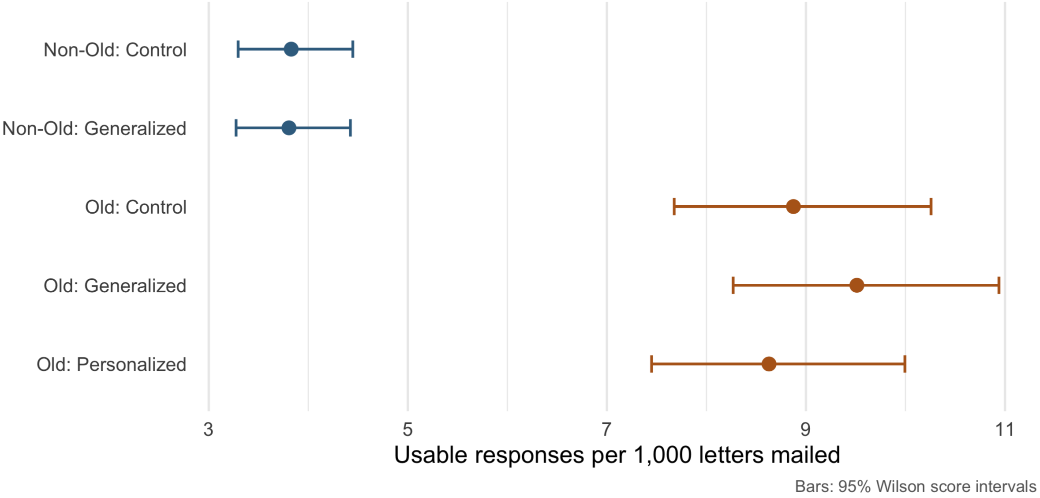
Realized Allocation

Based on the Round 1 Response Rate - Sent 120k letters in Round 2 (150k total) - Fewer disadvantaged than kits earmarked for them (all got a kit) - 100 kits in Round 1, 49 disadvantaged - 400 kits in Round 2, 224 disadvantaged

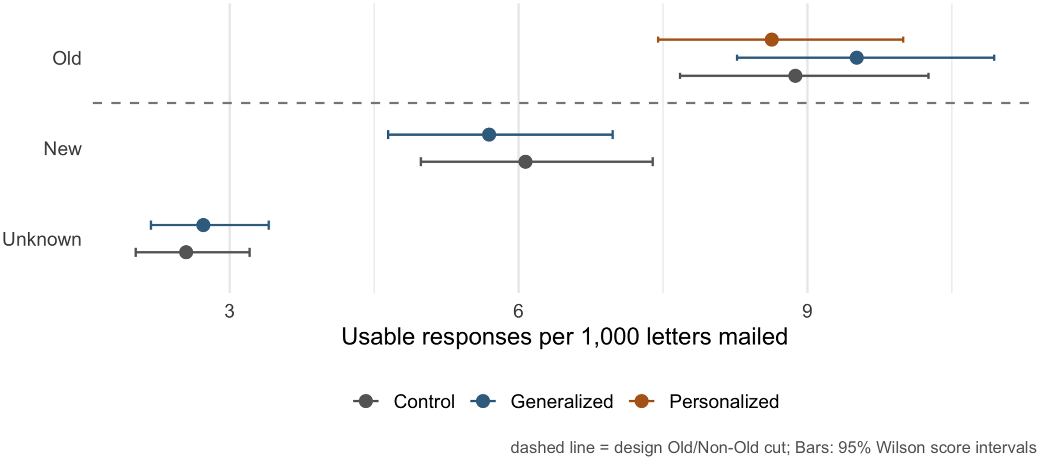
Description of outcomes

- Define “response” as response within 45 days of sending letter
- Define “usable” response as: GP, valid other stuff... (check this)
- How do we define “returning a blood kit”? We don’t have a deadline as in the intake survey, but at some point we stop the study!

Usable response rates vary with housing age not treatment.

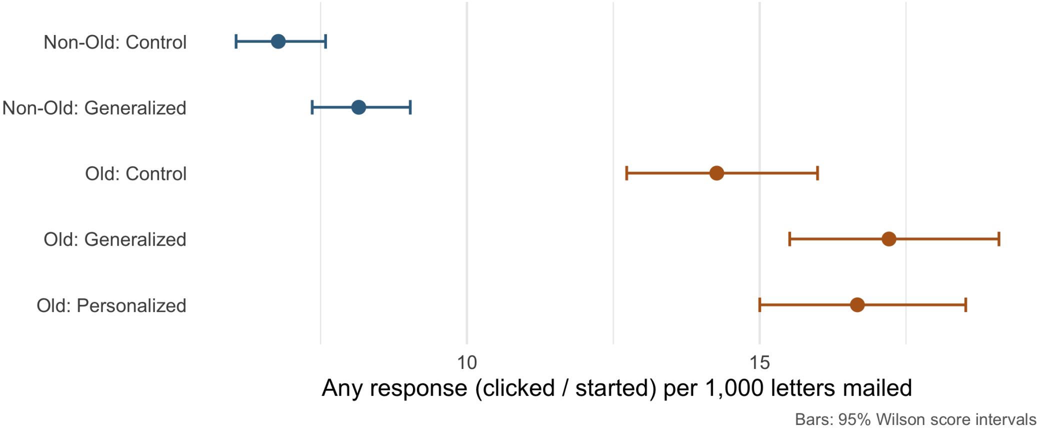


Highest response from Old (pre 1967); lowest from Unknown (no EPC)



Treatment affects *starting* the survey.

Preliminary — revisit after removing test rows.



Treatment effects *starting* the survey.

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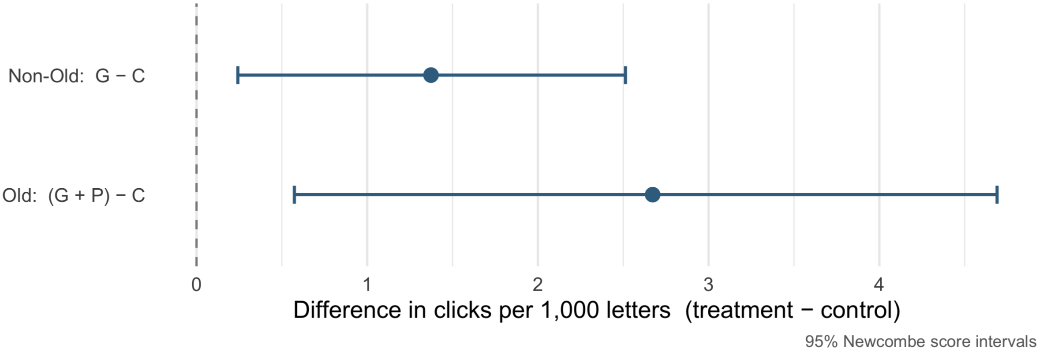
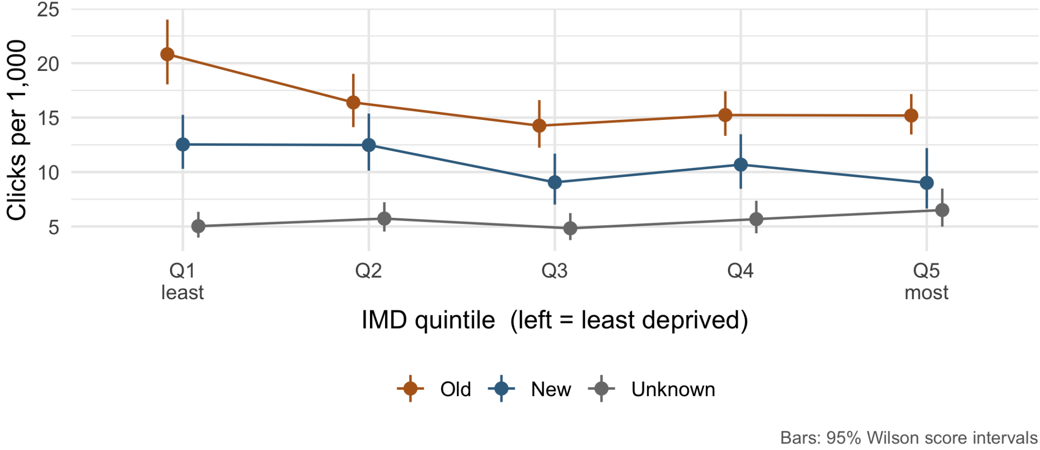


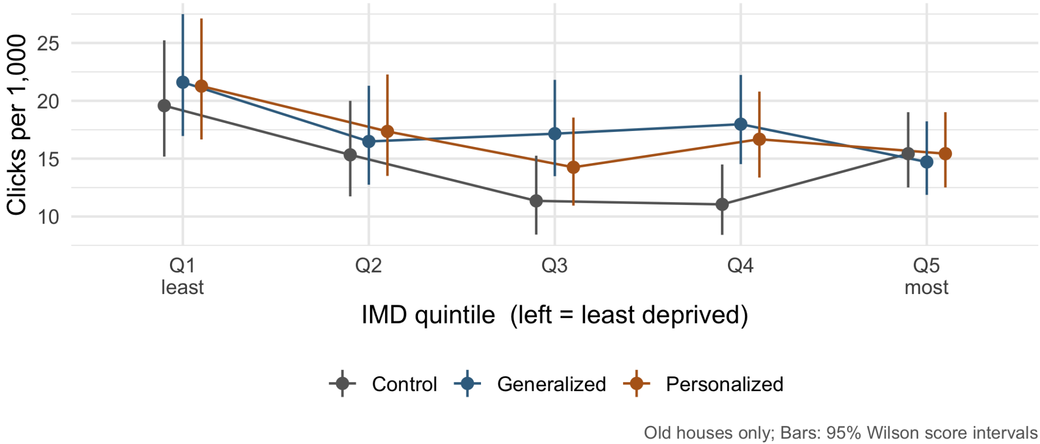
Clicking: deprivation gradient by housing age

Preliminary — revisit after test-row correction.



Clicking: does the gradient differ by treatment? (Old)

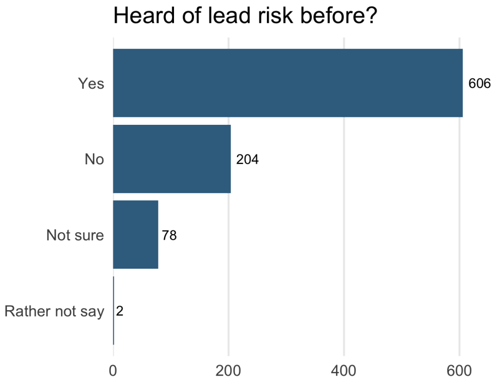
Preliminary — revisit after test-row correction.

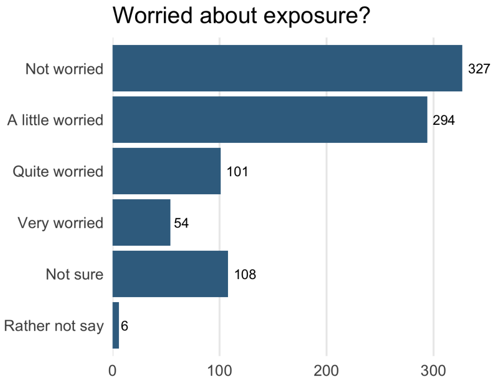


Intake Survey: Descriptive Statistics

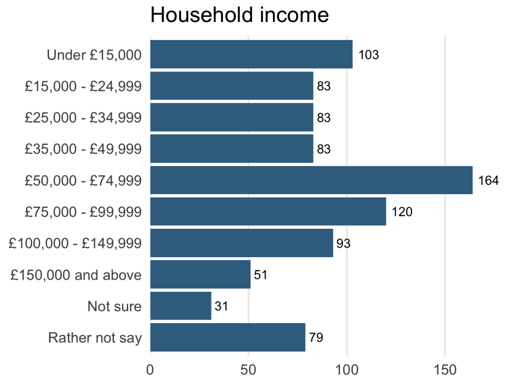
Intake Survey: Awareness of Lead

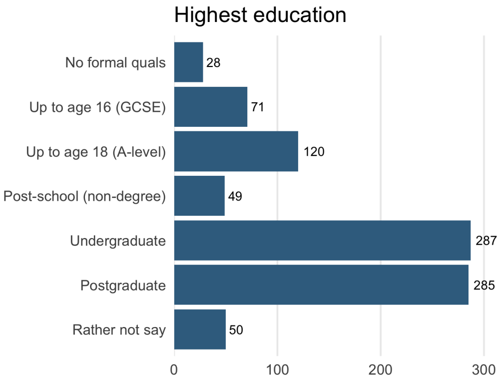
Usable respondents ($N = 890$); counts shown on bars.



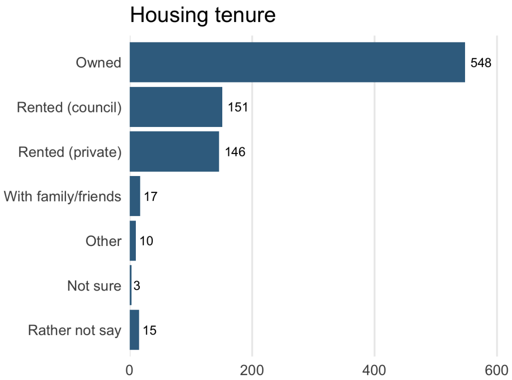


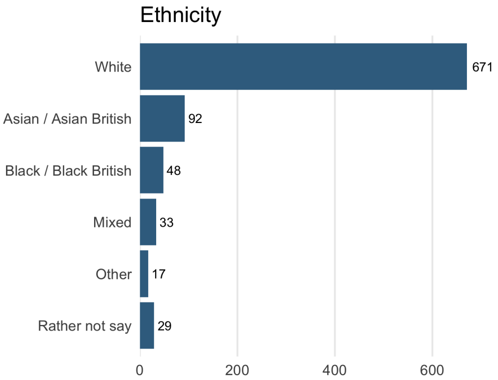
Intake Survey: Socioeconomic Profile



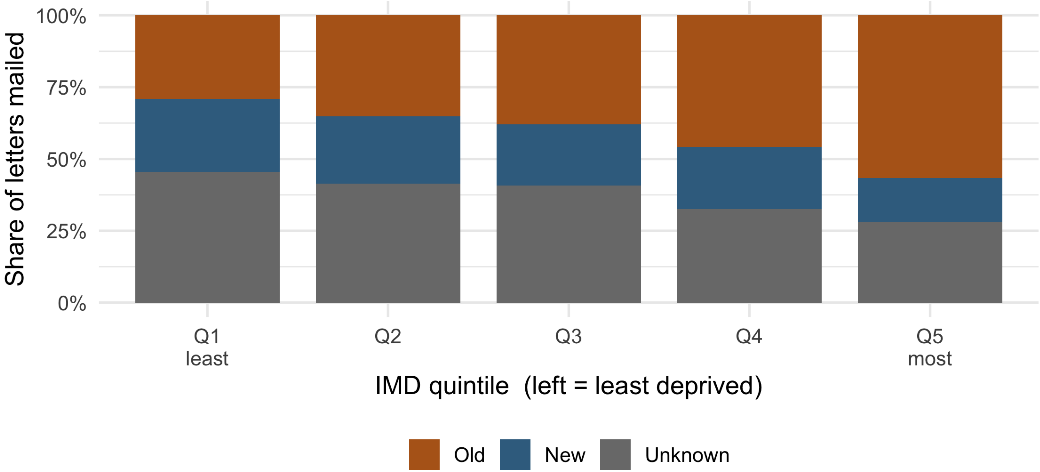


Intake Survey: Housing & Ethnicity





Old housing concentrates in deprived areas

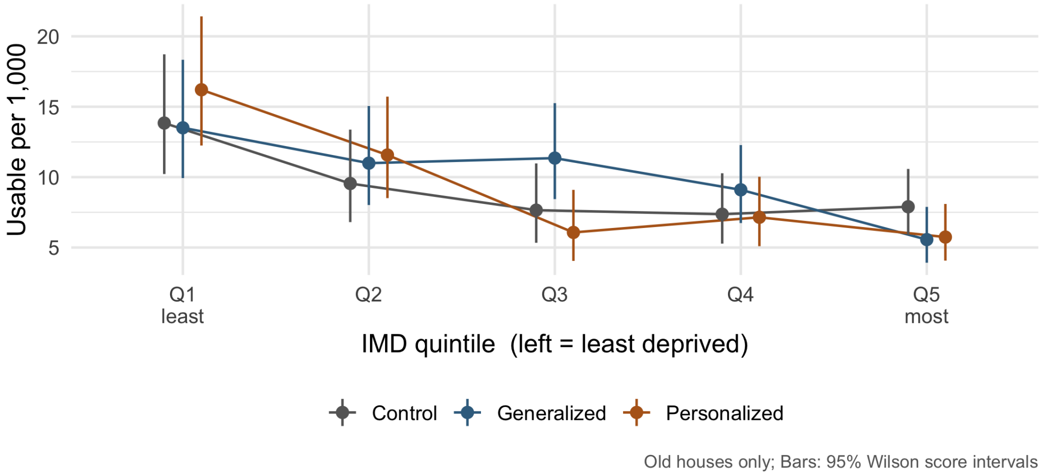


A puzzle to revisit

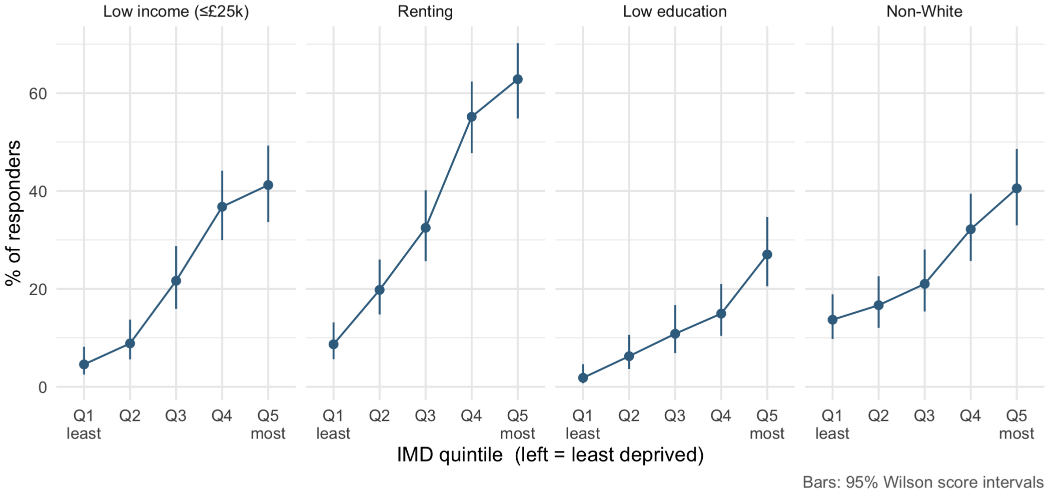
- Old houses respond **more** than New / Unknown.
- Old houses are **concentrated in the most-deprived areas**.
- Yet the most-deprived areas respond **less**.

How can all three be true at once? We come back to this.

Heterogeneity: does the gradient differ by treatment? (Old)



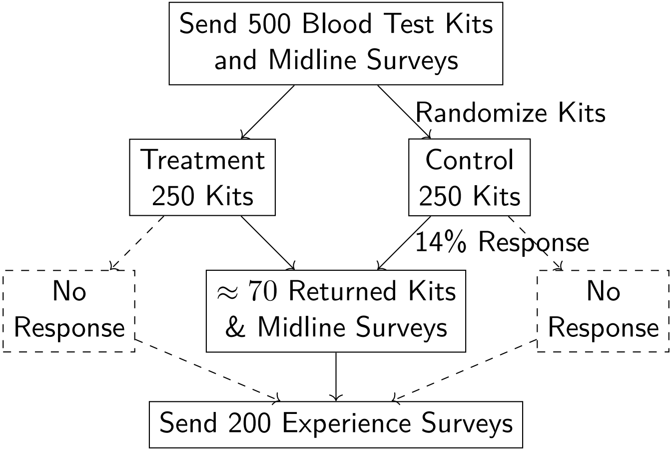
Ecological check: do responders' own traits track the neighbourhood?



Phase II: Collecting Blood Samples

Update numbers in this flowchart; can give current estimate of response rates although still accepting round 2 kits

Baseline survey & census: choose probs of sending each household a kit (2000 to 500)



Who returns a blood sample?

- Needs to be put into the repo in the anonymized form and merged.
- Talk to Andrea about this today!
- Consider adding some very preliminary results about who sends back a kit (valid kit?)
- Look at this by "golden envelope" treatment
- Could also correlate with demographics / knowledge about lead etc. These need to be re-weighted so they're slightly trickier. Use the re-balancing weights.

Who returns the exposure survey?

- Ask Andrea about this; she has sent the vouchers for this so we can probably modify her existing code to do this; should already be anonymized and ready-to-use?
- Can correlate this with the same things as returning a blood sample: kit type, demographics, knowledge about lead, etc.

Feedback survey

- It was opened a few days ago we think? Ask Andrea
- Perhaps we could present some very preliminary results?

Categorizing letters

- RA is still going through all the letters that were returned
- Hasn't finished yet
- We could add a slide here explaining what we plan to do with that information when we get it.
- E.g. we could try to correct the response rate estimates and we could check to make sure that this wasn't related to LSOA characteristics or we might find out that it is!

Appendix

Sampling Design: Exclusion / De-duplication

Addresses Excluded

- Hotels / student residences: 936
- Vacant properties: 931
- Missing UPRN: 73
- Missing LSOA: 524
- Missing postcode (after EPC fallback): 3491

De-duplication

- After exclusion, 18 UPRNs appear *exactly twice*
- These represent errors in the Council Tax data
- Only affects address; keep first one from each pair

Final Sampling Frame: 370,666 rows

- Address 1-4, postcode, UPRN
- OA (2,607), LSOA (488), MSOA (107)
- EPC Housing age: "Old" if ≤ 1966 (missing for 38.5%)
- $\#$ Households in LSOA $\geq$ one child aged ≤ 9 (2021 Census)